

ASSESSMENT 2: THEORY TEST 2 BRIEF

Subject Code and Title	CAI104 Concepts in AI
Assessment	Assessment 2: Workbook
Individual/Group	Individual
Length	Workbook
Learning Outcomes	This assessment addresses the Subject Learning Outcomes outlined at the bottom of this document.
Submission	Due by 11:55pm AEST Sunday end of Module 8.
Weighting	35%
Total Marks	100 marks

Task Summary

In this assessment, you will be given a workbook that includes a set of questions to complete. You have to answer and show your working for the questions in the workbook. This assessment should be done individually. Please refer to the Task Instructions for details on how to complete this task. The rubric at this end of this document will be used to mark your answers. The assessment is intended to test:

- Your understanding of the histories and theories covered in Modules 1 to 8 (inclusive).
- Your ability to use uninformed and informed search methods to solve simplified real-world problems.
- Your ability to use knowledge representation and reasoning methods to solve simplified real-world problems.

Context

With this assessment, you demonstrate your ability to tackle and solve several cases studies using AI search methods, knowledge representation techniques, AI theories, logics, and reasoning approaches that you have learned throughout Modules 1-8.

Task Instructions

To complete this assessment task you must:

- Download the Assessment 2 Workbook (question sheet) from the Assessment 2 Resource Folder in Blackboard.
- Make sure to read the questions carefully and understand them before you start.
- Answer **ALL** questions.
- Show your working and the steps taken for each answer. Provide examples for your discussions and answers to demonstrate your thorough understanding. Partial mark will be awarded for correct working even if the final answer is incorrect.

Submission Instructions

This assessment should be submitted online:

- Name and save your file as “CAI104_Assessment2_LastName_FirstName.pdf”
- Submit your file via the Assessment 2 Submission link found under the Assessment tab in the main navigation menu.
- You can see your learning facilitator’s feedback via the Grade Centre in the LMS portal (My Grades).

Assessment Rubric

Assessment Attributes	Fail (Yet to achieve minimum standard) 0-49%	Pass (Functional) 50-64%	Credit (Proficient) 65-74%	Distinction (Advanced) 75-84%	High Distinction (Exceptional) 85-100%
Work demonstrates knowledge and application of uninformed and informed search algorithms 30%	Little or no knowledge of uninformed and informed search algorithms. Most of the answers are wrong. There is no discussion on the applications and efficiency.	Acceptable but further work is required to show the knowledge of uninformed and informed search algorithms. Some of the answers are wrong, and discussions are not satisfactory.	Good level of knowledge about uninformed and informed search algorithms. The answers are correct, and the discussions show good but not highly relevant applications.	Excellent but not thorough knowledge about uninformed and informed search algorithms. The answers are correct. The discussions and examples are relevant but not in-depth.	Excellent and thorough understanding of uninformed and informed search algorithms. The answers are correct. The discussions are in-depth and applications are highly relevant.
Work demonstrates knowledge and application of knowledge representation methods 30%	Little or no knowledge of knowledge representation methods. Most of the answers are wrong. There is no discussion on the applications and efficiency.	Acceptable but further work is required to show the knowledge of knowledge representation methods. Some of the answers are wrong, and discussions are not satisfactory.	Good level of knowledge about knowledge representation methods. The answers are correct, and the discussions show good but not highly relevant applications.	Excellent but not thorough knowledge about knowledge representation methods. The answers are correct. The discussions and examples are relevant but not in-depth.	Excellent and thorough understanding of knowledge representation methods. The answers are correct. The discussions are in-depth and applications are highly relevant.
Work demonstrates knowledge and application of logics in AI 20%	Little or no knowledge of logics in AI. Most of the answers are wrong. There is no discussion on the applications and efficiency.	Acceptable but further work is required to show the knowledge of logics in AI. Some of the answers are wrong, and discussions are not satisfactory.	Good level of knowledge about logics in AI. The answers are correct, and the discussions show good but not highly relevant applications.	Excellent but not thorough knowledge about logics in AI. The answers are correct. The discussions and examples are relevant but not in-depth.	Excellent and thorough understanding of logics in AI. The answers are correct. The discussions are in-depth and applications are highly relevant.
Work demonstrates knowledge and application of reasoning approaches	Little or no knowledge of reasoning approaches. Most of the answers are wrong. There is no discussion on the	Acceptable but further work is required to show the knowledge of reasoning approaches. Some of the answers are wrong, and	Good level of knowledge about reasoning approaches. The answers are correct, and the discussions show good but	Excellent but not thorough knowledge about reasoning approaches. The answers are correct. The discussions and examples are relevant	Excellent and thorough understanding of reasoning approaches. The answers are correct. The discussions are in-depth and

20%	applications and efficiency.	discussions are not satisfactory.	not highly relevant applications.	but not in-depth.	applications are highly relevant.
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The following Subject Learning Outcomes are addressed in this assessment	
SLO a)	Demonstrate an understanding of the field of artificial intelligence including its origins, fundamental concepts, limitations and the problems that it tries to solve.
SLO b)	Explain basic concepts, methods, theories and application for search.
SLO c)	Interpret and formulate knowledge representations in the form of logic expressions.
SLO d)	Describe knowledge systems, the use of AI based problem solving methods, forms of knowledge representation, and model-based reasoning.